

Project Report

Project Name: CatCare

Reference ID: HBGN008N

1. Executive Summary

CatCare is a fully offline Android companion app for cat owners that centralises feeding schedules, health tracking, vet reminders, and gamified streaks in one place. It addresses the lack of simple, privacy-first cat care tools by requiring no internet connection, no account, and no third-party services — making it accessible to any cat owner instantly.

2. Project Overview

2a. Why you're building what you're building

Cat owners often juggle multiple reminders, vet records, and feeding times mentally or across different apps. There was no single offline-first app that combined all of these with a motivation layer (streaks/badges) to encourage consistent care habits. CatCare solves this with a clean, joyful experience purpose-built for cat parents.

2b. How it relates to the theme

CatCare directly serves two hackathon categories "For the Cats" (feeding schedules, health tracking, and symptom guide improve cat wellbeing) and "For Cat Owners" (vet reminders, weight charts, and streak gamification help humans be better cat parents). Every feature has a direct positive impact on cats and the people who care for them.

2c. Target Audience

Cat owners of all ages, especially those who want a simple, private, offline tool to track their cat's daily care without signing up for accounts or sharing personal data.

3. Key Features

List the major core features you built:

- ☐ Cat Profiles — add multiple cats with name, breed, age, weight, gender, and photo
- ☐ Feeding Schedule — set meal times per cat with daily local notification reminders
- ☐ Health Log — log checkups, symptoms, vaccines, and weight entries per cat
- ☐ Vet & Vaccine Reminders — date-based reminders with optional morning notifications
- ☐ Symptom Guide — 30+ offline cat symptoms with causes, actions, and urgency levels (searchable)
- ☐ Weight Chart — line graph of weight history with trend analysis (stable/gain/loss)

- ❑ Badges & Streaks — 8 unlockable badges + daily streak tracking to reward consistent care

4. Technology Stack

List the languages, frameworks, libraries, and platforms used. Layer is something like (frontend, backend, database, etc)

Layer	Technology
Language	JAVA
Platform	Android (min API 26 / Android 8.0)
IDE	Android Studio
UI Framework	Material Design 3 Components
Data Storage	SharedPreferences (JSON serialized)
Notifications	AlarmManager + BroadcastReceiver
Chart Library	MPAndroidChart v3.1.0
Image Library	CircleImageView v3.1.0
Build System	Gradle

5. Technical Architecture

CatCare uses a single-Activity architecture with fragment-based navigation via a bottom NavigationView. All data persistence is handled by a single **PrefsHelper** utility class that serializes each data type (cats, meals, health entries, reminders, streak, badges) as JSON arrays stored in Android's SharedPreferences.

Data flow:

User Action



Fragment / Activity



PrefsHelper (read/write)



SharedPreferences (on-device JSON)



UI Update (re-render list/chart)

Notifications flow:

User sets reminder time



AlarmManager schedules alarm



NotificationReceiver (BroadcastReceiver) fires



NotificationManager shows notification

Key components:

- **MainActivity** — hosts bottom nav, swaps fragments
- **PrefsHelper** — all CRUD operations + streak/badge logic
- **SymptomGuideActivity** — static data + live search filter
- **WeightChartActivity** — reads health entries, renders MPAndroidChart line graph with trend analysis
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6. Testing Matrix

Optional section, but this will improve your marks on technical execution. List the manual test cases you ran to verify your project works as intended.

Feature / Flow	Steps	Expected Result	Actual Result	Pass / Fail
Add cat	Fill form → Save Kitty	Cat appears on Home	Cat card shown	Pass
Edit cat	Tap cat card → Edit fields → Save	Updated info shown	Updated correctly	Pass
Delete cat	Edit screen → Delete → Confirm	Cat removed from list	Removed instantly	Pass
Add meal	Feeding tab → Add Meal → Save	Meal card appears	Peach card shown	Pass
Delete meal	Tap “  ” icon on meal card	Meal removed	Removed instantly	Pass

Feature / Flow	Steps	Expected Result	Actual Result	Pass / Fail
Feeding notification	Set meal time 2 mins ahead, enable reminder	Notification fires at set time	Notification received	Pass
Add health entry	Health tab → Add Entry → Save	Entry shown, streak updates	Entry + streak updated	Pass
Weight chart	Add 3+ health entries → Open chart	Line graph renders	Graph with dots shown	Pass
Trend analysis	Add entries with increasing weight	Shows gain warning	Correct trend shown	Pass
Add reminder	Reminders tab → Add → Pick date → Save	Lavender card appears	Card shown sorted	Pass
Mark reminder done	Tap checkbox	Strikethrough on title	Strike shown	Pass
Symptom search	Type "vomit" in search	Filtered results only	Correct filter	Pass
Badge unlock	Add first cat	First Kitty badge earned	Badge shown	Pass
Streak increment	Log health entry on consecutive days	Streak +1	Streak updated	Pass
Offline test	Enable airplane mode → use app	Full functionality	All features work	Pass
No cats state	Open Feeding with no cats	"Add a cat first" message	Empty state shown	Pass

8. Future Improvements

If you had another week, what would you build or improve next? Show the judges you thought beyond the deadline.

- ☐ Multi-cat comparison charts (weight trends side by side)
- ☐ Cat vaccination history export as PDF
- ☐ Community lost & found cat map (stray cat reporting)
- ☐ Breed-specific care tips and feeding recommendations
- ☐ Apple iOS version using Flutter

9. Tools You Used

List any tools that helped you build the project (e.g. AI coding assistants, design tools, recording software, etc.)

- ☐ Android Studio (IDE)
- ☐ Claude AI (coding assistance and architecture guidance)
- ☐ Material Design 3 (UI component system)
- ☐ MPAndroidChart (open source chart library)
- ☐ GitHub (version control)
- ☐ Aikido Security (security scanning)
- ☐ Canva / DaVinci Resolve (demo video editing)

11. Learnings & Takeaways

What did the team learn, technically or otherwise, from building this project?

- ★ Building fully offline apps forces cleaner architecture — no network error states to handle
- ★ SharedPreferences with JSON is surprisingly capable for structured local data without Room DB complexity
- ★ Gamification (streaks + badges) requires careful state management across multiple user actions
- ★ Material Design 3 dramatically reduces time spent on UI polish
- ★ A clear README and structured documentation is as important as the code itself for judging

12. Acknowledgments

Optional: credit any mentors, open-source projects, APIs, or datasets that helped along the way.

MPAndroidChart by Philipp Jahoda (open source, Apache 2.0 license)

CircleImageView by Henning Dodenhof (open source)

Material Design Components by Google

Cat health symptom information referenced from general veterinary guidance

Submission Checklist

- ☐ Video demo (HD or at least 720p)
- ☐ README.md (prerequisites, run instructions, configuration)
- ☐ Project report (this document, in documentation/)
- ☐ Source code in src/

- ☐ No unrelated files, executables, auto-generated code, or package folders (e.g. node_modules/, build/, dist/, vendor/)